

Answer Key

Question 1

1. B
2. A
3. D
4. D
5. C
6. A

Question 2

(a) $N = M/L = 10 \times 10^6 / 2000 = 5000$ packets.

(b)

Transmission delays:

$$t_1 = 2000 / (2 \times 10^6) = 1 \text{ ms.}$$

$$t_2 = 2000 / (1 \times 10^6) = 2 \text{ ms.}$$

Propagation delays: $d_1 = 20 \text{ ms}$, $d_2 = 30 \text{ ms}$.

First packet delay: $(1 + 20) + (2 + 30) = 53 \text{ ms}$.

Bottleneck link: $t_2 = 2 \text{ ms}$.

Remaining packets: $(5000 - 1) \times 2 = 9998 \text{ ms}$.

Total delay: $53 + 9998 = 10051 \text{ ms} \approx 10.051 \text{ s}$.

Question 3

(a) Non-persistent HTTP/1.0 (no parallel):

Transmission for Base = $20 \times 10^3 \times 8 / (10 \times 10^6) = 16 \text{ ms}$.

Transmission for each image = $50 \times 10^3 \times 8 / (10 \times 10^6) = 40 \text{ ms}$.

Base HTML: $2(40) + 16 = 96 \text{ ms}$.

8 images: $8 \times [2(40) + 40] = 960 \text{ ms}$.

Total: $96 + 960 = 1056 \text{ ms} \approx 1.056 \text{ s}$.

(b) Non-persistent HTTP/1.0 (5 parallel connections):

Base HTML: 96 ms.

Batch 1 (5 images): $2\text{RTT} + \text{transmission} = 80 + 40 = 120 \text{ ms}$.

Batch 2 (remaining 3 images): 120 ms.

Total: $96 + 120 + 120 = 336 \text{ ms} \approx 0.336 \text{ s}$.

(c) Persistent HTTP/1.1 with pipelining:

TCP handshake (1 RTT = 40 ms) + GET base HTML (1 RTT + transmission = 40 + 16 = 56 ms) + Browser pipelines all 8 image GETs simultaneously (Total transmission for all 8 images = 8×40 ms = 320 ms + 1 RTT = 40 + 320 = 360 ms).
Total: 40 + 56 + 360 = 456 ms.

Question 4

- (a) 5000, 6000, 7000
 (b) 6000
 (c) $\text{ssthresh} = \text{cwnd} / 2 = 20 \text{ KB} / 2 = 10 \text{ KB}$, $\text{cwnd} = 1 \text{ KB}$
 (d) $\text{ssthresh} = \text{cwnd} / 2 = 20 \text{ KB} / 2 = 10 \text{ KB}$, $\text{cwnd} = \text{ssthresh} + 3 \text{ MSS} = 10 \text{ KB} + 3 \text{ KB} = 13 \text{ KB}$
 (e) 1 KB
 (f) 1 KB
 (g) 5 further duplicate ACKs.

Question 5

- (a)
- i) Subnet A Address: 200.10.128.0/22
 - ii) Subnet A Broadcast: 200.10.131.255/22
 - iii) Subnet B Address: 200.10.132.0/23
 - iv) Subnet B Broadcast: 200.10.133.255/23
 - v) Subnet C Address: 200.10.134.0/25
 - vi) Subnet C Broadcast: 200.10.134.127/25

(b) Forwarding Table:

Destination Prefix	Interface ID
* (Default)	3
200.10.128.0/22	0
200.10.132.0/23	1
200.10.134.0/25	2

Question 6

(a) Dijkstra's Algorithm:

Step	N'	D(v), P(v)	D(w), P(w)	D(x), P(x)	D(y), P(y)	D(z), P(z)
0	u	1, u	∞	4, u	2, u	∞
1	uv	-	6, v	4, u	2, u	∞
2	uvy	-	5, y	4, u or 4, y	-	4, y
3	uvyx	-	5, y	-	-	4, y
4	uvyxz	-	5, y or 5, z	-	-	-
5	uvyxyzw	-	-	-	-	-

(b) Forwarding Table for router u:

Destination	Outgoing Link	Cost
V	(u, v)	1
W	(u, y) or (u, v)	5
X	(u, x)	4
Y	(u, y) or (u, v)	2
Z	(u, y) or (u, v)	4

Question 7

(a) Three (3)

(b) Count to Infinity Problem. Solution is called Poison Reverse or Split Horizon.

(c)

i) $5 - 1 = 4$ bits

ii) Zero (0)

iii) Better / stronger error detection by CRC than Checksum, as checksum may miss errors, specially burst errors.

Question 8

(a) Yes. A generates an ARP request for 10.0.0.2 & B replies.

(b) Yes.

(c)

Switch Table: S1

MAC Address	Port
A	1
B	2
C	3

Switch Table: S2

MAC Address	Port
C	1
A	3
B	3